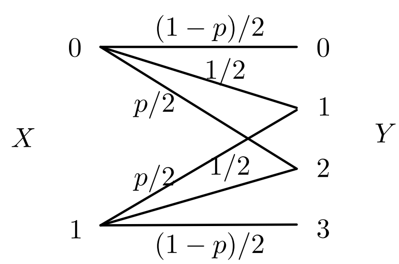
COMELEC department UE COM105

Exercise 6

<u>Goal of the exercise :</u> We study the capacity of two systems.

Exercise 1

We consider the discrete memoryless channel with the following channel diagram :



where p is a real-valued constant between 0 and 1.

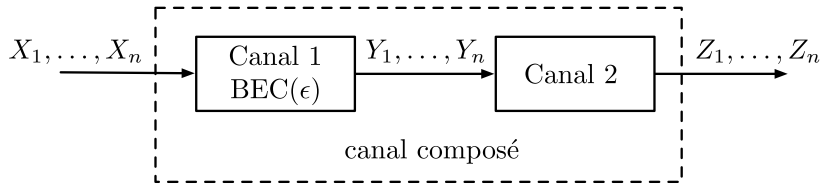
Questions :

1. Let $p = 0$. For this special case, redraw the channel diagram and compute the capacity.
2. Let $p = 1$. For this special case, redraw the channel diagram and compute the capacity.
3. We compute the channel capacity for an arbitrary $p \in [0, 1]$. To this end, we consider X following a Bernoulli-q distribution for an arbitrary $q \in [0, 1]$.
   (a) Find $H(Y|X = 0)$, $H(Y|X = 1)$, and $H(Y|X)$.
   (b) We introduce the random variable $Z = \mathbb{1}\{Y \in \{0, 3\}\}$, where $\mathbb{1}\{\cdot\}$ is the indicator function that equals 1 if the condition is satisfied and 0 otherwise. Show that
   $$H(Y) = H(Z) + H(Y|Z).$$
   (c) Find $H(Z)$. Does this entropy depend on the parameter q and thus on the distribution of X ?
   (d) Find the maximum value of the conditional entropy $H(Y|Z)$ over all parameters $q \in [0, 1]$.
   (e) By combining the answers from questions (a)–(d), determine the channel capacity for $p \in [0, 1]$.

Exercise 2 on the back.

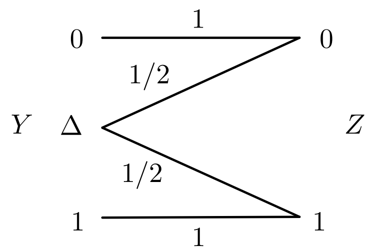
Exercise 2

We consider a sequence of two memoryless channels (DMC) as shown in the following figure :



The output symbols of channel 1, $Y_1, \ldots, Y_n$, thus form the input symbols of channel 2. We can then define a composite channel that takes the input symbols, $X_1, \ldots, X_n$, and produces the output symbols, $Z_1, \ldots, Z_n$. It can be verified that this channel is again a DMC.

For the first channel, we consider a BEC with an erasure probability $\epsilon \in [0, 1]$. The second channel is given by the following channel diagram, where Δ is the erasure symbol of channel 1.



Questions :

1. Find the diagram of the composite channel with input X and output Z. Is this channel familiar to you ?
2. Show that the capacity of the composite channel cannot be greater than the capacity of channel 1 by following these steps :

   Hint : For any triplet of random variables (A, B, C), we define the mutual information between A and the pair (B, C) as $I(A; B, C) := H(A) - H(A|B, C)$, where

   $$H(A|B,C) := \sum_{b,c} P_{B,C}(b,c) \sum_a P_{A|B,C}(a|b,c) \log_2 \frac{1}{P_{A|B,C}(a|b,c)}.$$

   It can be shown that $H(A|B, C) \leq H(A|B)$.

   (a) Show that $H(X|Y, Z) = H(X|Y)$, when X is the input of the first channel, Y its output and at the same time also the input of channel 2, and Z the output of channel 2.
   (b) Show that $I(X; Y) = I(X; Y, Z)$.
   (c) Show that $I(X; Y, Z) \geq I(X; Z)$.
   (d) Show that the capacity of channel 1 cannot be smaller than the capacity of the composite channel.
3. What do you conclude about the capacities of BSC and BEC channels ?